

Online Appendix to

“AI Exposure or Occupational Composition? Constructed Measures and Early-Career Employment”

Lily Luo

September 2026

Contents

A	Exposure architectures and measurement lineage	3
A.1	Dependence and candidate census	5
A.2	What a validation sample would require	5
B	Calendar, mapping, and age-allocation uncertainty	8
C	Residual-treatment and conditional-information support	11
C.1	Support after broad-family conditioning	12
D	Estimator, precision, and time-dependence audits	12
D.1	Estimating equation and maintained randomness	12
D.2	Canonical interval and detectable effects	13
D.3	Sparse cells and frequency	13
D.4	Rotation and cross-occupation time covariance	14
D.5	Pre-AI pseudo-breaks	14
E	Employment-stock composition and comparison audits	16
F	Dynamics and population-control diagnostics	18
F.1	Event-time specification	18
F.2	January 2025 population controls	20
G	Longitudinal flow estimands and results	21
H	Mobility disagreement and an algorithm-defined benchmark	21

H.1	Pairwise disagreement and task-family redundancy	22
H.2	Identical support, omitted-support bounds, and uncertainty	22
H.3	Reversals do not identify a coding-error matrix	23
I	Continuous family models and exact reparameterization	24
J	Fixed-treatment occupation influence	26
K	Prospective precision, realized resolution, and failed diagnostics	28
L	Reproducibility and design chronology	28

A Exposure architectures and measurement lineage

The six architectures differ before any labor outcome enters. Three versions of the ability-based index alter the weighting used to aggregate occupation–ability links. Three task-based measures use the direct-exposure share (alpha), the direct plus complement-software share (beta), and the broad direct-or-complement share. The source files, transformations, and occupation-level checks are recorded in the replication ledger. Figure 1 summarizes the construction paths.

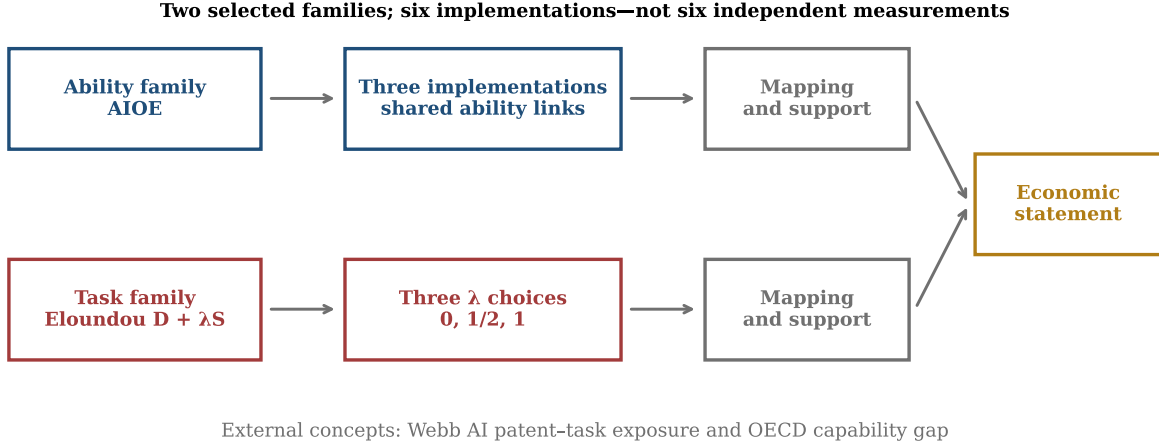


FIGURE 1. Measurement lineage across selected and external AI-exposure architectures. Each path ends in an occupation-level score before merger to the employment data. Ability-based, task-based, patent-overlap, and capability-gap inputs represent distinct treatment constructions, not alternative labels for observed adoption.

TABLE 1. Employment-weighted correlations among exposure architectures

	AIOE-A	AIOE-D	AIOE-S	Alpha	Beta	Broad
<i>Panel A. Pearson correlations</i>						
AIOE administrative	1.000	0.984	0.996	0.295	0.833	0.890
AIOE ability-direct	0.984	1.000	0.981	0.276	0.821	0.883
AIOE source-weighted	0.996	0.981	1.000	0.293	0.832	0.890
Alpha	0.295	0.276	0.293	1.000	0.643	0.346
Beta	0.833	0.821	0.832	0.643	1.000	0.941
Broad	0.890	0.883	0.890	0.346	0.941	1.000
<i>Panel B. Average-rank correlations</i>						
AIOE administrative	1.000	0.987	0.993	0.274	0.844	0.874
AIOE ability-direct	0.987	1.000	0.982	0.258	0.845	0.880
AIOE source-weighted	0.993	0.982	1.000	0.269	0.843	0.874
Alpha	0.274	0.258	0.269	1.000	0.567	0.369
Beta	0.844	0.845	0.843	0.567	1.000	0.961
Broad	0.874	0.880	0.874	0.369	0.961	1.000

Notes: Frozen treatment-side reference sample of 463 occupations, weighted by pre-period employment. Panel B is the weighted Pearson correlation of deterministic average ranks. These are descriptive treatment-side comparisons.

TABLE 2. Employment-weighted correlations with occupational characteristics

Architecture	Cognitive	Manual	RTI	Educ.	Wage	Remote	STEM	Computer
AIOE administrative	.653	-.936	.161	.688	.611	.741	.249	.849
AIOE ability-direct	.689	-.913	.152	.708	.640	.746	.242	.847
AIOE source-weighted	.660	-.939	.148	.700	.617	.754	.249	.854
Alpha	-.032	-.266	.217	-.034	.011	.200	.436	.304
Beta	.478	-.758	.169	.425	.478	.589	.371	.797
Broad	.598	-.810	.108	.530	.600	.618	.253	.833

Notes: Descriptive Pearson correlations using frozen pre-period employment weights. RTI is routine-task intensity. These characteristics are not validation labels for true AI exposure.

For the task family, the raw primitives satisfy $X_\alpha = D$, $X_\beta = D + S/2$, and $X_\gamma = D + S$. On strict common support, the maximum absolute gap between raw beta and $(\alpha + \gamma)/2$ is 7.0×10^{-13} . Mapping and aggregation preserve the identity. Separate employment-weighted standardization changes the relative location and scale and creates a maximum numerical midpoint gap of 0.5112; it does not create a third primitive.

The external-architecture admission rule was written before fitting external outcomes. Webb AI and the reversed OECD capability gap meet the declared construct, public-source, non-title mapping, full-component, distinct-cut, and 80-percent coverage criteria. Webb software remains a computerization control; its source correlation with Webb AI is 0.7021. Frey–Osborne remains an automation-risk comparator, not an AI architecture.

A.1 Dependence and candidate census

TABLE 3. Weighted principal-component spectrum of the six core implementations

Component	Variance share	Cumulative share
PC1	80.04%	80.04%
PC2	16.07%	96.11%
PC3	3.47%	99.58%
Remaining three	0.42%	100.00%

Notes: Standardized scores on the exact 444-occupation six-score-plus-Webb common support, weighted by preperiod employment. Principal components summarize dependence and are not admitted as additional architectures or latent truth.

A.2 What a validation sample would require

Generated-covariate corrections require information absent here: labels that are closer to the target construct than the indices being evaluated. A useful validation design would sample occupation–task units across all major occupation families and the full distribution of architecture disagreement; oversample split-route occupations and tail-boundary cases; and separately record (i) whether a frontier model can complete the task, (ii) realized time savings with and without complementary software, (iii) actual workplace use, and (iv) task importance and frequency. The distinction prevents capability, complementarity, adoption, and economic weight from being collapsed into one label.

Labels should come from blinded independent domain experts and, where feasible, audited use logs or randomized tool access. Raters should not see the candidate score, occupation outcome, or treatment group. Repeated labels permit inter-rater reliability and adjudication;

TABLE 4. Repository-bounded candidate architecture census

Candidate set	Status	Primitive/family	Reason or limitation
Three AIOE implementations	Core, selected	Application–ability	Same construct system; later external-admission rule not applied retroactively
Eloundou alpha, beta, broad	Core, selected	LLM task-time rubric	Two primitives D, S ; three implementations are dependent
Webb AI	External pass	Patent–task text overlap	Public, hashed, mapped, complete, distinct cuts, adequate coverage
OECD capability gap	External pass	Nine-domain capability/demand gap	Same operational pass; different horizon and construct
Frey–Osborne	Scope failure	Broad automation risk	Instantiated comparator but not relabeled as AI exposure
Yin/Vu/Persico, Yin/Ogut, Budget Lab, BCC alternatives, retrieval/RL/query families	Pending instantiation	Several primitives	Primary source noted, but no candidate-specific hashed score, route, support, and admission record
Firm- and demand-side referee candidates	Unit match/unverified locally	mis- Firm or labor-demand measures	Cannot be inserted into occupation CPS without a defensible employer/industry bridge

Notes: This table classifies the verified repository record; it is not a claim to enumerate all published AI measures. A pass means an operational screen was met, not construct validity. Future versions should report source integrity, construct scope, mapping/support, and representation feasibility in separate columns.

a held-out set of occupations tests transport across families and taxonomies. The validation release must preserve task text/version, model version and date, tool configuration, rater protocol, occupation crosswalk, and sampling probabilities.

There is no design-free universal sample size. For example, under independent occupation units, a Fisher- z calculation for distinguishing validation correlations .4 and .6 with 80-percent power at a two-sided 5-percent level requires about 112 occupations:

$$n \simeq 3 + \left\{ \frac{1.96 + .842}{\operatorname{atanh}(.6) - \operatorname{atanh}(.4)} \right\}^2 .$$

Task clustering, unequal sampling, rater noise, missing logs, and the desired precision of a corrected downstream coefficient all increase that number. A serious design would report power over assumed prevalence, reliability, intraclass correlation, and holdout share rather than advertise one threshold. The resulting labels could support reliability correction, validation-sample moments, or generated-regressor inference; the current data cannot.

B Calendar, mapping, and age-allocation uncertainty

The original extract selected March ASEC (‘03s’) rather than Basic Monthly (‘03b’) samples in 2017–2021. ASEC records carry zero Basic Monthly final weights for this design. A separately authenticated repair restores the five months. October 2025 is genuinely absent because the CPS was not collected. Combined files contain 9,843,021 rows and 114 observed survey months; the corrected static model retains 113 after omitting December 2022.

TABLE 5. Calendar and taxonomy sensitivities

Specification	Months	Occ.	Coefficient	95% interval
Frozen chronology benchmark	108	468	-.1311	[-.2171,-.0451]
Restored March 2017–21	113	468	-.1346	[-.2223,-.0468]
No possible split inbound route, fixed labels	113	369	-.1674	[-.2684,-.0663]
No possible split inbound route, recomputed labels	113	369	-.1471	[-.2428,-.0513]
Stable Census-2010 aggregate	113	408	-.1522	[-.2438,-.0606]
Respondent-equivalent cells	113	468	-.1348	[-.2216,-.0481]

Notes: The first row is confirmatory; the others are post-outcome exploratory. The two one-to-one-target rows and stable-taxonomy row change the occupational population. Respondent equivalents reconstruct cells without CPS final weights; they are not weighted stocks divided by a constant.

Pre-2020 records use official Census-2010-to-2018 routes. Common conversion proportions are applied to young and older source records, mechanically preserving a source age ratio across its allowed targets before other sources enter. Split sources account for 14.93 percent of young and 17.75 percent of older 2017–19 stock on primary support. In Q1 the shares are 17.48 and 17.13 percent; in Q5 they are 21.57 and 26.32 percent.

Ninety-nine primary targets can receive a split source and carry 23.38 percent of fitted Q5 target information. Allocating information in proportion to split-source stock assigns 4.44 percent to that stock, but the allocation is descriptive rather than identified. From 2020 onward CPS occupations are directly coded in the target Census-2018 taxonomy, so this bridge uncertainty affects the preperiod rather than postperiod observations.

Official routes contain no age-specific allocation probabilities. Unrestricted allocation within allowed routes produces sharp 2017–19 stock-accounting ranges of [.1031,.1480] for the Q1 young/older ratio and [.0790,.1175] for Q5. These are not coefficient bounds. A predeclared exposure-directed tilt changes young-versus-older odds toward the highest-beta eligible target by $K \in \{.5, .75, 1, 1.5, 2\}$ and keeps labels fixed. The coefficient ranges from $-.1292$ to $-.1391$. Only 20 split sources and 6.88 percent of matched early stock are eligible, so the exercise cannot establish global bridge robustness.

TABLE 6. Reconciliation of occupation universes

Universe	Occupations	Months used to form support	Availability rule
Raw route-expanded balanced candidates	539	66	Positive young and older preperiod stock
Frozen beta-complete support	490	66	Raw balance plus finite beta
Repaired AIOE plus Webb support	495	66	Raw balance plus finite repaired AIOE and Webb
Intersection of preceding two supports	466	66	Satisfies both architecture-specific rules
Primary beta plus Webb model	468	108/113	Finite beta and Webb in frozen/corrected static panel
Six core measures plus Webb	444	108	Literal complete-case intersection

Notes: The 490- and 495-occupation supports are not nested: 24 occupations occur only in beta-complete support and 29 only in repaired-AIOE-plus-Webb support. The first four rows define preperiod availability; the final two are fitted-model samples after the full calendar and model-specific requirements. Frozen and corrected calendars contain 108 and 113 static-model months, respectively.

The AIOE compatibility gate fails closed when a SOC-2010 source is matched literally to Census-2018 outcome codes without a versioned route. Naive support with Webb is 410 occupations, 85 fewer than the repaired 495. Computer/mathematical coverage rises from 3.33 to 97.7 percent after repair. On fixed 410-occupation support, changing the score moves a continuous coefficient only from $-.01885$ to $-.01920$; expanding to 495 moves it to $-.03156$. This isolates implementation correction from population re-admission.

C Residual-treatment and conditional-information support

Support depends on the statement being estimated. A continuous horse race consumes residual score variation after controls; a tail comparison consumes weighted group membership; a ranking statement consumes pairwise order. The diagnostics below are therefore not generic data-coverage statistics.

For the primary beta tail contrast, estimator information is concentrated more sharply than the nominal count suggests. The frozen information audit reports an effective occupation count of approximately 43.3 for the fitted Q5-versus-Q1 contrast, compared with 468 nominal occupations. The pre-outcome residual-treatment audit reports an effective count of 53.263. These objects have different weights and should not be equated.

The computerization comparison uses Webb’s software-exposure measure as the frozen primary comparison technology. Archived O*NET computer-use measures, routine-task intensity, and Frey–Osborne automation risk are sensitivities. Remotability is a separate proxy and not a measure of computerization. The comparison-technology results show that coefficient magnitude depends on the conditioning estimand; they do not select a uniquely correct exposure measure.

TABLE 7. Effective residual support across comparison technologies

Architecture	Webb	O*NET importance	O*NET level	RTI	Frey–Osborne
AIOE administrative	72.2	73.0	77.3	62.8	63.8
AIOE ability-direct	74.6	77.8	82.2	68.2	59.3
AIOE source-weighted	72.1	65.4	70.3	62.1	62.8
Alpha	17.4	31.1	31.7	11.9	26.3
Beta	53.3	63.2	59.5	51.7	62.9
Broad	84.5	44.4	72.2	77.5	74.0

Notes: Inverse-Herfindahl counts of employment-weighted residual-exposure variance after projecting each exposure on the named comparison technology. Frozen pre-outcome treatment-side analysis; no labor outcome enters.

Top-five residual-variance shares for the same columns are 18.4, 17.9, 16.6, 20.5, and 20.0 percent for administrative AIOE; 15.9, 16.5, 15.0, 17.8, and 20.6 for ability-direct AIOE; 18.1, 19.5, 18.3, 20.6, and 20.2 for source-weighted AIOE; 41.6, 32.8, 33.3, 46.6, and 36.0 for alpha; 22.2, 20.1, 20.6, 22.8, and 19.8 for beta; and 15.8, 25.2, 18.0, 17.3, and 17.1 for broad exposure.

C.1 Support after broad-family conditioning

The repaired baseline uses 468 occupations in 22 SOC2 groups. Every group contains at least two beta quintiles, but only four contain both Q1 and Q5. Thus a Q5–Q1 coefficient can still be estimated after SOC2 conditioning through a connected set of intermediate-quintile comparisons, even though most families do not supply a direct tail-to-tail contrast.

Conditional target information falls from 26.44 million in the baseline to 7.95 million after adding SOC2-by-post terms and 7.93 million after absorbing SOC2-by-month paths. The corresponding weighted residual SD falls from .158 to .087. Effective-information counts rise from 43.36 to about 55.8 because the remaining information is less concentrated, but this should not be mistaken for more total identifying information: only 30 percent of the baseline information remains. The estimated coefficient consequently has an SE around .0705 rather than .0450.

The original six-score stable tails supply a different support diagnostic. Forty-six occupations are always Q1 and 18 always Q5, together representing 9.74 percent of employment. The stable-tail estimate has effective information 15.02 and a top-five share of 46.7 percent. Common occupational coverage therefore does not imply common classification or diffuse estimator support.

D Estimator, precision, and time-dependence audits

D.1 Estimating equation and maintained randomness

For two age groups, conditioning Poisson means (μ_y, μ_o) on their weighted total yields the grouped-binomial pseudo-log-likelihood $N_y \log p + N_o \log(1 - p)$ with $\text{logit}(p) = \log(\mu_y/\mu_o)$. Occupation-month effects cancel. CPS final weights construct N_y and N_o and are not applied again. These continuous stocks define an estimating equation rather than literal independent binomial trials.

The primary sandwich and wild-score interval treat occupation score contributions as independent clusters and allow arbitrary serial dependence within each occupation. They condition on the observed cells, treatment labels, mapping, support, weights, and model. They do not propagate the CPS complex-survey design, exposure-label generation, or taxonomy-allocation uncertainty. In the nuisance-adjusted implementation, regressors are absorbed against occupation and calendar-month effects under fitted binomial information weights. Scores are summed within occupation and premultiplied by the inverse information matrix. A $\sqrt{G/(G - 1)}$ finite-cluster correction is applied. Each bootstrap draw multiplies the whole

occupation influence vector by one Rademacher sign; the analytic cluster SE is the studentizer. Paired analyses reuse the same occupation signs.

D.2 Canonical interval and detectable effects

TABLE 8. Canonical primary interval and precision

Calendar	Estimate	Cluster SE	95% wild-score interval	p	MDE ₈₀
Frozen 108 months	-.131074	.044410	[-.217075, -.045073]	.0030	.124418
Repaired 113 months	-.134554	.044957	[-.222307, -.046801]	.0016	.125950

Notes: Intervals use 9,999 draws. MDE₈₀ is normal-theory, two-sided, at the 5-percent level. The repaired row is the post-outcome substantive baseline; the frozen row remains the confirmatory design record.

Earlier drafts attached four slightly different intervals to the frozen coefficient because separate paired exercises generated different sets of 999 multipliers. Those finite-draw intervals ranged from $[-.2190, -.0431]$ to $[-.2166, -.0455]$. A minimum-count row also used a different 463-occupation sample. The table above supplies the sole canonical primary interval; pair-specific member intervals are labeled and are not substituted elsewhere.

All seven beta-minus-alternative Q5–Q1 intervals include zero. Paired MDE₈₀ values are .1272, .1250, .1273, .1053, .0609, .1499, and .1689 for the three AIOE implementations, alpha, broad, Webb AI, and OECD, respectively. The inference permits “does not detect a difference,” never economic equivalence.

D.3 Sparse cells and frequency

TABLE 9. Respondent equivalents per occupation-month-age cell

Age group	p10	Median	p90	Zero	Below five
Ages 22–25	0	2	16	26.32%	69.93%
Ages 26–65	4	25	199	2.05%	11.07%

Notes: There are 101,088 age-specific cells on the frozen support. One-sided boundary cells remain valid grouped-binomial contributions. Only the 965 occupation-months with both age stocks zero are omitted.

Selecting cells with at least five observed young respondent equivalents retains 46.6 percent of occupation-months and gives -0.1090 (interval $[-0.2050, -0.0131]$). Requiring five in both groups gives -0.1097 . These are outcome-count-selected diagnostics, not cleaner primary estimates. Quarterly aggregation gives -0.13103 on the frozen calendar and -0.13454 on

the repaired calendar, with SEs .04441 and .04495. Monthly sparsity therefore does not mechanically generate the point estimate.

D.4 Rotation and cross-occupation time covariance

TABLE 10. Cross-occupation time-score covariance sensitivity

Time-HAC lag	SE	Normal 95% interval	MDE ₈₀
0	.04493	[-.21914, -.04300]	.12589
1	.04655	[-.22232, -.03983]	.13042
4	.04830	[-.22575, -.03640]	.13533
12	.04909	[-.22729, -.03485]	.13754
16	.04897	[-.22705, -.03510]	.13719

Notes: The sensitivity adds a Newey–West covariance of aggregate monthly scores and subtracts the occupation-month intersection. It is model-based, not a CPS replicate-weight estimator. Occupation clustering already permits arbitrary within-occupation serial dependence.

The planning simulation omitted several dependencies and predicted SE .01217. Replacing independent occupation signs with one global sign per path preserves the donor month’s full contemporaneous cross-occupation residual vector. Across all 132 enumerated offset/sign paths, the mean model SE remains .01217 and the empirical SD is .00659. This sharp sensitivity closes none of the gap to .04441. It rejects one candidate mechanism, not all forms of cross-occupation dependence.

D.5 Pre-AI pseudo-breaks

The extract begins in 2017, so 2015–16 pseudo-breaks cannot be run. In the 12 balanced 2017–19 dates, the median is approximately $-.0015$, the 5th percentile $-.0064$, and the 95th percentile .0128. None is as negative as $-.1311$. The plus-one empirical tail is therefore $1/13 = .0769$, the smallest attainable value in this dependent set, not a conventional p -value. The more extreme estimates among all 34 feasible dates occur at the beginning of the window with as little as one pre-break month.

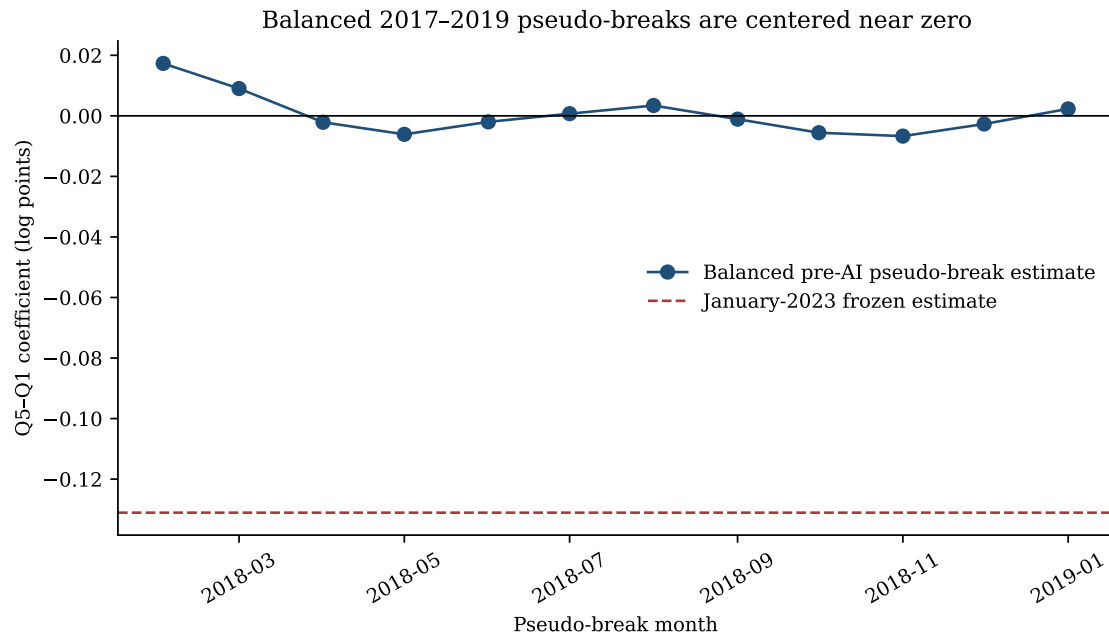


FIGURE 2. Balanced pre-AI pseudo-break distribution

Notes: The 12 dates retain at least 12 months on each side and construct exposure groups using only 2017-2019 weights. The dashed line is the frozen January-2023 estimate. Overlapping break estimates are dependent; the post-outcome exercise is descriptive rather than a randomization test.

E Employment-stock composition and comparison audits

All exercises in this section are post-outcome exploratory. They preserve the frozen artifacts as a benchmark and use 9,999 occupation-level wild-score draws unless stated otherwise.

TABLE 11. Broad-family conditioning

Calendar/model	Coef.	SE	95% interval	Information	Effective occ.	Top-five share
Frozen baseline	-.1311	.0444	[-.2171,-.0451]	25.68m	43.30	.246
Frozen, SOC2 $\times$ post	-.0278	.0712	[-.1650,.1095]	7.73m	55.60	.211
Frozen, SOC2 $\times$ month	-.0281	.0709	[-.1646,.1085]	7.70m	55.83	.211
Repaired baseline	-.1346	.0450	[-.2223,-.0468]	26.44m	43.36	.245
Repaired, SOC2 $\times$ post	-.0315	.0706	[-.1676,.1046]	7.95m	55.69	.211
Repaired, SOC2 $\times$ month	-.0317	.0703	[-.1674,.1040]	7.93m	55.92	.211

Notes: All six fits are full rank. The two baseline rows use their canonical 9,999-draw intervals; conditioned rows use the composition audit’s common-draw intervals. SOC2 conditioning changes the estimand and leaves approximately 30 percent of baseline target information. All 22 broad groups contain at least two quintiles, but only four contain both Q1 and Q5.

On the repaired calendar, the paired difference of SOC2-by-post minus baseline is .1031 (paired SE .0511, interval [.0035,.2026], $p = .042$). SOC2-by-month minus baseline is .1028 (paired SE .0509, interval [.0035,.2021], $p = .042$). These direct differences establish detectable coefficient movement; they are not a causal share decomposition.

TABLE 12. Quintile profile and reference contrasts

Contrast	Estimate	SE	95% interval	MDE ₈₀
Q5–Q1	-.1311	.0444	[-.2171,-.0451]	.1244
Q5–Q2	-.0456	.0406	[-.1248,.0335]	.1137
Q5–Q3	-.0833	.0387	[-.1581,-.0084]	.1081
Q5–Q4	-.0340	.0393	[-.1110,.0429]	.1099
Q4–Q2	-.0116	.0316	[-.0728,.0496]	.0884

Notes: The fitted profile relative to Q1 is $[0, -.0855, -.0478, -.0970, -.1311]$. The joint Q2–Q5 equality test has $p = .119$; the one-sided max- t monotonicity assessment has $p = .393$. Neither equality nor monotonicity is established.

The always-Q1/always-Q5 population contains 64 occupations and 9.74 percent of common-

support employment. Its coefficient is $-.2120$ (interval $[-.4131, -.0109]$), with effective information 15.02 and top-five share 46.7 percent.

TABLE 13. Influence and occupation-side service exclusions

Diagnostic	Occ. removed	Stock removed	Coef.	95% interval
Jointly delete five largest absolute LOCO movements	5	4.11%	-.1011	[-.1755,-.0267]
Jointly delete ten	10	10.00%	-.1522	[-.2377,-.0667]
Jointly delete twenty	20	14.33%	-.1555	[-.2296,-.0814]
Symmetric 2.5% signed-influence trim	24	15.67%	-.1281	[-.1993,-.0568]
Huber down-weight above p95 absolute movement	0	—	-.1314	[-.1996,-.0632]
Drop all food preparation/service (SOC35)	10	3.27%	-.1201	[-.1993,-.0409]
Drop Q1 food preparation/service	5	1.17%	-.1196	[-.1988,-.0404]
Drop all SOC35/37/39 services	33	8.20%	-.1370	[-.2217,-.0523]
Drop Q1 SOC35/37/39 services	17	4.89%	-.1387	[-.2236,-.0538]

Notes: The LOCO ranking is selected using prior outcome influence, so joint deletion, trimming, and down-weighting are diagnostic rather than preferred estimators. Service definitions use Census-2018 occupation major groups and frozen treatment assignments. They do not isolate reopening or labor-supply mechanisms.

TABLE 14. Public CPS bridge to the BCC exposure grouping

Architecture, native support	Occ.	Coef.	95% interval	MDE ₈₀
AIOE administrative	495	-.0532	[-.1104,.0039]	.0828
AIOE ability/direct	484	-.0679	[-.1268,-.0090]	.0853
AIOE OEWS-weighted	485	-.0610	[-.1184,-.0035]	.0827
Eloundou alpha	468	-.0741	[-.1250,-.0232]	.0722
Eloundou beta	468	-.0728	[-.1240,-.0216]	.0728
Eloundou broad	468	-.0817	[-.1342,-.0292]	.0748
Webb AI	448	-.0168	[-.0710,.0374]	.0784
OECD capability gap	448	-.0096	[-.0600,.0407]	.0728

Notes: Each architecture forms its own employment-weighted top-two and bottom-three groups. Native support mixes construction and support. On literal 426-occupation common support, only beta minus OECD is detected: $-.0733$, paired interval $[-.1390, -.0076]$. This is a different contrast from Q5–Q1.

F Dynamics and population-control diagnostics

F.1 Event-time specification

The event-study model replaces the single Q5-by-young-by-post coefficient with month-specific Q5-by-young interactions, retaining occupation-by-age, occupation-by-month, age-by-month effects, Q2–Q4 post terms, and the Webb software interaction. October 2022 is the coefficient reference month. December 2022 remains in the event-study data as the declared product-launch transition month but is not assigned to the static post indicator; the static model omits it. October 2025 is absent because no CPS was collected. The distinction between a reference coefficient and a transition-period rule resolves the earlier caption ambiguity.

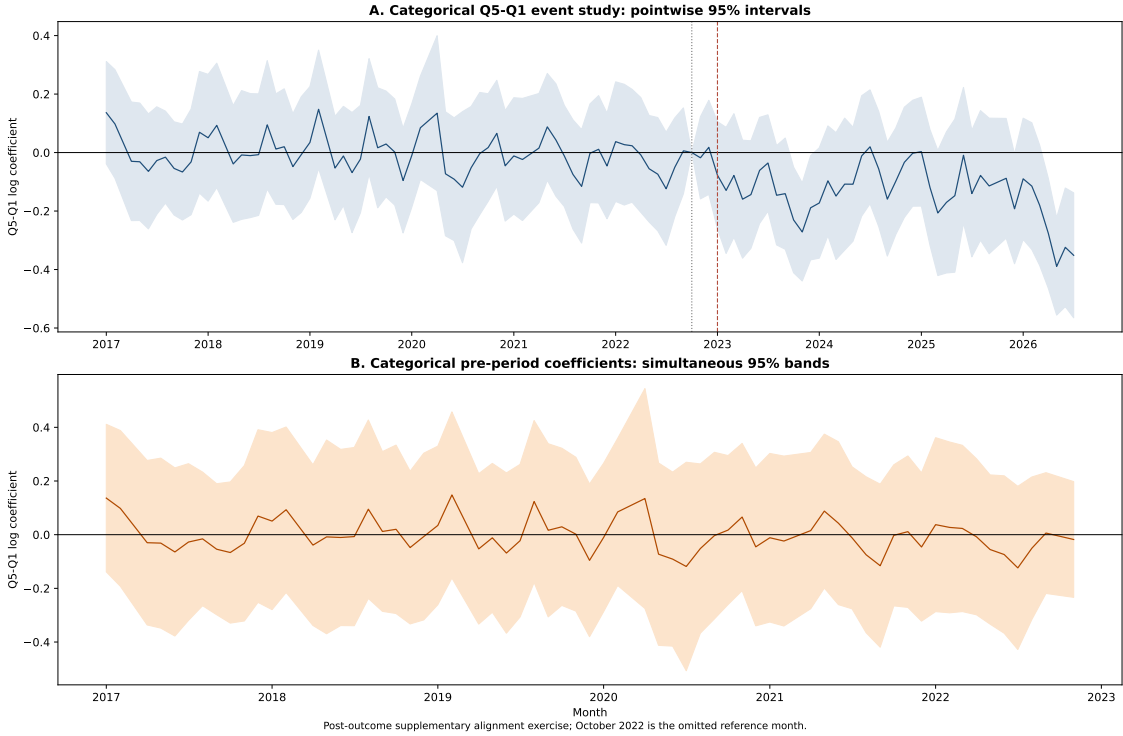


FIGURE 3. Categorical beta Q5–Q1 event-time profile

Notes: October 2022 is the omitted coefficient reference. December 2022 is the transition month for the static before/after specification. The model is the frozen categorical event study and the figure is descriptive; it does not measure adoption.

The preperiod max statistic is 1.502 with common-multiplier $p = .929$, so the implemented model does not detect a differential pretrend. That test does not establish parallel trends or AI specificity. Forty of 42 post coefficients are negative, but the path is neither immediate nor monotone. The balanced pre-AI pseudo-break analysis appears in Appendix D.

The post-2025 extension differs from 2024 but not from 2023. Interest rates, technology-

TABLE 15. Joint post-period coefficients

Period or contrast	Coefficient	Paired SE	95% interval
2023	-.1356	.0452	[-.2245,-.0467]
2024	-.0783	.0548	[-.1821,.0254]
2025–26	-.1629	.0507	[-.2623,-.0635]
2024 minus 2023	.0573	.0447	[-.0302,.1447]
2025–26 minus 2023	-.0273	.0440	[-.1154,.0608]
2025–26 minus 2024	-.0846	.0374	[-.1577,-.0115]

sector adjustment, return-to-office changes, post-pandemic normalization, and composition remain concurrent explanations.

F.2 January 2025 population controls

BLS introduced Vintage-2024 population controls in January 2025, raising the estimated civilian noninstitutional population by approximately 2.9 million and employment by approximately 2 million. Official prior months were not revised. BLS’s experimental bridge is aggregate and supplies no age-by-occupation adjustment, so a literal subgroup counterfactual weight series is unavailable.

TABLE 16. Weight and endpoint sensitivities

Cell construction/window	Coefficient	SE	95% interval	Months
Official weights, repaired full window	-.1346	.0450	[-.2226,-.0465]	113
Official weights, ending December 2024	-.1135	.0451	[-.2009,-.0261]	95
Respondent equivalents, repaired full window	-.1348	.0441	[-.2216,-.0481]	113
Respondent equivalents, ending December 2024	-.1056	.0465	[-.1968,-.0143]	95

In a joint official-weight model, 2023–24 is $-.1108$ and 2025–26 $-.1664$; the paired difference is $-.0556$ with interval $[-.1228, .0116]$. Under respondent equivalents the difference is $-.0800$ with interval $[-.1390, -.0211]$. The raw official-weight Q5-minus-Q1 log ratio becomes $.0139$ less negative from December 2024 to January 2025, whereas the respondent series becomes $.0302$ more negative. These diagnostics do not recover a missing weight vintage, but the official revision does not mechanically account for the full postperiod pattern.

G Longitudinal flow estimands and results

All flow models use valid exact adjacent-month CPSIDV links, origin age, official origin LNKFW1MWT, January 2017–November 2022 as pre, January 2023 onward as post, and December 2022 as transition. Links across the MISH 4-to-5 rotation gap are excluded. December 2019 origins are excluded from occupation-dependent transitions.

For exit, the risk set is employed origins and the event is nonemployment at $t + 1$. For occupational outflow, the risk set is employed origins linked to an employed destination and the event is a different harmonized occupation. For entry destination, the risk set is nonemployed origins linked to employment; the outcome is allocation toward destination Q5 rather than Q1. Consequently, the three coefficients are not additive decompositions of the stock coefficient.

TABLE 17. Longitudinal flow estimates

Margin	Coefficient	Analytic SE	Wild CI	p
Employment exit	0.1195	0.0944	$[-0.0668, 0.3059]$	.229
Occupational outflow	0.0107	0.0631	$[-0.1063, 0.1277]$	.857
Persistent outflow	-0.0246	0.1195	$[-0.2580, 0.2088]$	.850
Entry destination	-0.0888	0.0976	$[-0.2787, 0.1011]$	.383

Notes: Post-outcome exploratory conditional-Poisson rate or allocation models using the frozen beta Q5-versus-Q1 contrast and official longitudinal weights. Inference uses 999 occupation-cluster wild-score draws. Each interval includes zero. Entry destination is an allocation estimand, not an employment-finding probability.

Dependent interviewing does not eliminate occupation coding error. The earlier 9.865-percent immediate-reversal statistic comes from a broader ages-18–65 feasibility universe. On the six-architecture ages-22–65 support used in Appendix H, 6,187 $A \rightarrow B \rightarrow A$ observations are 10.336 percent of switches with an observable third interview. Either denominator conditions on a third adjacent interview, and neither statistic is a measured coding-error rate or a basis for correcting the stock coefficient.

H Mobility disagreement and an algorithm-defined benchmark

The mobility exercises are post-outcome exploratory diagnostics of constructed classifications. They use 108,500 employed-to-employed switches on six-way common support and official

longitudinal weights. They neither estimate a hiring margin nor validate a latent occupation code.

H.1 Pairwise disagreement and task-family redundancy

TABLE 18. Pairwise zero-threshold directional disagreement

	AIOE-A	AIOE-D	AIOE-S	Alpha	Beta	Broad
AIOE-A	—	8.64	3.42	39.88	26.56	21.76
AIOE-D	8.64	—	10.63	42.59	27.72	21.54
AIOE-S	3.42	10.63	—	40.22	25.30	21.22
Alpha	39.88	42.59	40.22	—	23.46	34.82
Beta	26.56	27.72	25.30	23.46	—	11.26
Broad	21.76	21.54	21.22	34.82	11.26	—

Notes: Percent of official switch weight. A pair disagrees when its two strictly nonzero score movements have opposite signs; exact zeros are ties. Block means are 7.56 percent within AIOE, 23.18 percent within task shares, and 29.64 percent between families. Giving each block one-third weight produces a 20.13-percent family-balanced summary. This normalization is descriptive, not a welfare metric.

The six-way probability that at least one of the six dependent implementations opposes another is 53.28 percent. It is not a typical pairwise disagreement rate. The task measures also satisfy the exact raw-score identity $\beta = (\alpha + \text{broad})/2$, with maximum numerical error 7.0×10^{-13} . Thus β cannot introduce a new zero-threshold sign conflict when α and broad agree; positive standardization preserves this result. Across 66,582 same-endpoint-sign switches there are zero violations.

TABLE 19. Six-way mobility conflict by standardized threshold

Threshold	Eligible weight	Any conflict eligible	Opposition/all	Opposition eligible
0	100.00	53.28	53.28	53.28
.10	99.78	53.18	43.35	43.45
.25	97.42	52.78	29.79	30.58
.50	85.29	50.99	14.00	16.41

Notes: Percent. “Any conflict” requires one positive and one negative movement after restricting to switches where at least one architecture exceeds the threshold. “Opposition” requires both directions to exceed it. Standard-deviation thresholds are statistical decision rules, not economic-loss or welfare scales.

H.2 Identical support, omitted-support bounds, and uncertainty

Hamilton allocation at 200,000 pseudo-units represents 84,192 of 94,893 detailed transition cells, 97,759 of 108,500 switch rows, and 98.305 percent of official weight. Recomputing both

sides on that identical represented support gives the following audit.

TABLE 20. No-self rematching benchmark on identical support

Object	Estimate	Uncertainty measure
Represented-support realized conflict	53.2819%	household-cluster SE 0.1979 pp
Sealed no-self benchmark	52.3227%	Monte Carlo SE 0.00172 pp
Realized minus benchmark	0.9592 pp	sampling SE 0.0776 pp

Notes: The realized-rate percentile interval is [52.8947,53.6591] percent. The gap’s interval, centered on the sealed support-aligned point and using the Monte Carlo-variance-subtracted sampling SE, is [0.8070,1.1113] percentage points. Its raw cluster-plus-simulation SE is 0.0857 pp. The empirical diagnostic uses 399 mean-one exponential multipliers over 74,671 longitudinal households; each replicate rebuilds a 200,000-unit plug-in benchmark with two rematches. It is not CPS replicate-weight or full complex-survey inference.

If conflict on the omitted 1.695 percent of benchmark support is unrestricted in $[0, 1]$, the all-support benchmark lies in [51.4359,53.1307] percent and the all-support realized-minus-benchmark gap lies in [0.1521,1.8469] percentage points. The sign survives this accounting bound, but the economic magnitude is not pinned near one point.

The sealed rule begins with a uniform destination permutation within each age-by-month-by-broad-origin-by-broad-destination stratum, repairs the first self-match using a feasible random swap, and restarts after an impasse. It preserves detailed margins and removes self-transitions, but it is not uniform over feasible derangements. An alternative that chooses the bad self-match uniformly before choosing a feasible partner gives 52.3211 percent, versus 52.3227 percent under the sealed rule; the difference is -0.00157 percentage points. This is one repair-rule sensitivity, not proof of a unique random- matching distribution. The defensible comparison is therefore “relative to the specified broad-assortative no-self benchmark,” not a causal explanation by broad assortativity.

H.3 Reversals do not identify a coding-error matrix

The earlier 9.865-percent reversal statistic used 11,121 A–B–A sequences among 112,736 observable first switches in a broader ages-18–65 feasibility universe. In the present ages-22–65 six-way support, 6,187 reversals are 10.336 percent of 59,858 switches with an observable third interview (10.663 percent under official weights), or 5.948 percent of official weight over all 108,500 switches. An A–B–A sequence may reflect a real temporary move, an assignment change, proxy reporting, or coding error. It does not reveal latent true occupation or a five-category error matrix. No attenuation curve is therefore constructed: under unknown differential error, Q5–Q1 misclassification may attenuate, amplify, or reassign the contrast.

I Continuous family models and exact reparameterization

This section is supplementary. The main text gives priority to the directly interpretable D, S and AIOE-plus-beta specifications. Crossing a significance threshold after deleting one component is not treated as a statistically significant coefficient change unless the direct paired change is tested.

Let A and E be equal-weight centroids of the three standardized ability- and task-family implementations. Raw $F = (A + E)/2$ and $G = (A - E)/2$ are separately standardized in the fitted model with Webb. The original construction has weighted SDs .8777 and .3221 and weighted correlation .2177; neither raw construction nor standardization makes the fitted coordinates orthogonal.

Equal weighting of standardized task measures is not neutral in primitive units. With weighted raw-score standard deviations $\sigma_\alpha = .14616$, $\sigma_\beta = .20455$, and $\sigma_\gamma = .33481$, the task centroid is, up to a centering constant,

$$E = \frac{1}{3} (\sigma_\alpha^{-1} + \sigma_\beta^{-1} + \sigma_\gamma^{-1}) D + \frac{1}{3} (\frac{1}{2}\sigma_\beta^{-1} + \sigma_\gamma^{-1}) S = 4.9057D + 1.8104S.$$

The embedded S -to- D weight is therefore .3690 in raw task-share units. This is an arithmetic consequence of component scales, not an economically optimal complementarity weight. Deleting a task-family component changes these effective weights rather than withdrawing an independent signal.

TABLE 21. Leave-one-measure F/G outcome refits

Omitted measure	b_F	95% interval	b_G	95% interval
None (original)	-.0404	[-.0674,-.0133]	.0309	[.0029,.0589]
Alpha	-.0330	[-.0571,-.0088]	.0342	[.0089,.0596]
AIOE administrative	-.0404	[-.0676,-.0133]	.0294	[.0009,.0580]
AIOE ability-direct	-.0401	[-.0672,-.0129]	.0338	[.0059,.0618]
AIOE source-weighted	-.0404	[-.0676,-.0133]	.0293	[.0015,.0572]
Beta	-.0421	[-.0702,-.0141]	.0287	[-.0009,.0584]
Broad	-.0383	[-.0652,-.0114]	.0215	[-.0080,.0511]

Notes: Identical 444-occupation support and family rebalancing after each omission. Each row refits the labor outcome with 999 common occupation multipliers. Results are post-outcome exploratory.

A representative two-score model estimates administrative AIOE .0449 (interval [.0012,.0885]) and beta $-.0788$ (interval $[-.1292, -.0284]$). The direct primitive model estimates $D = -.0323$ (interval $[-.0617, -.0030]$) and $S = \gamma - \alpha = -.0303$ (interval

$[-.0562, -.0044])$, conditional on Webb.

For separately scaled regressors, the exact basis transformation is

$$\gamma_A = \frac{1}{2}(b_F/s_F + b_G/s_G), \quad \gamma_E = \frac{1}{2}(b_F/s_F - b_G/s_G).$$

Consistently transforming every draw gives $\gamma_A = .0248$ (paired SE .0199) and $\gamma_E = -.0707$ (paired SE .0272). This is the same fitted predictor, not corroboration. A smaller variance alone does not mechanically create a larger standardized coefficient; covariance with the residualized outcome and with other regressors also matters.

For $D + \lambda S$, the categorical Q5–Q1 estimate is $-.1013$ at $\lambda = 0$, $-.1256$ at .25, $-.1297$ at .5, $-.1432$ at .75, and $-.1482$ at 1. Continuous slopes differ under fixed beta scaling and restandardization, so both conventions are stored. These are descriptions of conditional coefficients and coordinate choices, not a universal family ordering. In particular, the F/G significance pattern cannot establish that one family is correct or turn the implied A/E coefficients into independent outcome evidence.

J Fixed-treatment occupation influence

The leave-one-occupation-out audit deletes each occupation while retaining frozen treatment assignments and refits the same model. It is an influence diagnostic, not a redefinition of treatment. The primary full-sample estimate is -0.131074 ; deletion estimates range from -0.14238 to -0.11055 . The largest absolute movement is 0.02052 , so “no sign reversal” is not an adequate summary: one occupation moves the headline by 15.7 percent of its magnitude.

Fast food and counter workers are in frozen Q1; deleting them moves the estimate to -0.110553 (movement $+0.020521$). Customer service representatives are in Q5; deleting them moves the estimate to -0.113112 (movement $+0.017962$). Software developers are in Q5 and rank tenth by absolute influence; deleting them moves the estimate to -0.138787 (movement -0.007713). Thus, the result is not a software-developer artifact, while influential contributions arise at both exposure tails.

TABLE 22. Ten largest absolute occupation-deletion movements

Rank	Deleted occupation	Delete estimate	Movement
1	Fast food and counter workers	-0.11055	$+0.02052$
2	Customer service representatives	-0.11311	$+0.01796$
3	Home health aides	-0.14238	-0.01131
4	Computer occupations, all other	-0.12080	$+0.01028$
5	Office clerks, general	-0.14109	-0.01002
6	Secretaries and administrative assistants ^a	-0.14089	-0.00982
7	Food preparation workers	-0.13981	-0.00874
8	Laborers and freight/material movers, hand	-0.13946	-0.00839
9	Wholesale and manufacturing sales representatives	-0.13924	-0.00817
10	Software developers	-0.13879	-0.00771

Notes: Post-outcome fixed-treatment influence audit. The complete machine-readable table contains all 468 deletion estimates. Employment-stock-weighted signed-movement percentiles (1st, 5th, median, 95th, 99th) are -0.01002 , -0.00817 , -0.00001 , 0.00377 , and 0.01796 .

a Except legal, medical, and executive.

For the supplementary family-disagreement coefficient, all 444 deletion estimates remain positive between 0.025128 and 0.035993 . This fixed-treatment stability does not confer independent economic meaning on the G coordinate and is not promoted in the main text.

TABLE 23. Joint deletion and robust-influence diagnostics

Diagnostic	Coefficient	95% interval	Stock deleted
Delete top five by absolute LOCO movement	-.1011	[-.1755,-.0267]	4.11%
Delete top ten	-.1522	[-.2377,-.0667]	10.00%
Delete top twenty	-.1555	[-.2296,-.0814]	14.33%
Trim 2.5% from each signed-influence tail	-.1281	[-.1993,-.0568]	15.67%
Huber down-weight above p95 absolute deviation	-.1314	[-.1996,-.0632]	0

Notes: The occupation ranking and trimming rules use previously observed outcome influence, so all rows are post-outcome, data-adaptive diagnostics. The reversal between deleting five and deleting ten shows that influential occupations contribute with opposing signs; sequential deletions are not additive.

K Prospective precision, realized resolution, and failed diagnostics

The prospective primary SE was .01217 and the realized occupation-cluster SE is .04441, a ratio of 3.649. For beta minus alpha, .01167 versus .03759 gives a ratio of approximately 3.22 on the updated paired fit. The frozen paired design had 80-percent power to detect coefficient differences of approximately 3.27 percentage points under its planning DGP. No verified published estimate matched the YAX age groups, stock estimand, Q5–Q1 contrast, and scale, so binding equivalence inference was retired rather than assigned an arbitrary smallest effect of interest.

At realized precision, the primary normal-theory MDE_{80} is .1244 log points. Architecture-pair MDEs range from .0609 to .1689. Those values belong beside the architecture comparisons because they explain why point-estimate variation and statistical distinguishability are different claims.

The original historical simulation preserved observed exposure, weights, design matrices, and ordered residual paths. It omitted survey sampling, new postperiod shocks, structural composition change, exposure measurement uncertainty, and several dependence structures. A new sharp sensitivity preserves an entire donor month’s contemporaneous cross-occupation residual vector by applying one global sign. All 66 cyclic donor offsets and both signs yield 132 paths. The mean analytic SE is .012168, essentially identical to .012170 under the independent-sign simulation, while the empirical SD is .006591. The sensitivity closes none of the gap to .044410. It rules out this global-sign implementation as the explanation; it neither validates the original power calculation nor establishes that other common shocks are irrelevant.

Two failed diagnostics remain part of the record. The occupation-specific age-season model did not converge with 5,148 additional nuisance parameters. A literal residual-wild full-refit attempt produced no admissible grouped-binomial pseudo-outcome in 199 draws because transformed successes left $[0, N]$. That particular pseudo-outcome construction is inappropriate; its failure neither invalidates nor validates the one-step score bootstrap. No clipping or outcome-favorable substitute was introduced.

L Reproducibility and design chronology

The design-freeze tag resolves to commit `22fbf7924809b7a535e31ae0ab68f5b113ce8078`; the confirmatory-results tag resolves to `b16109482c3bf5ca176f6f08976e120b04769945`. These hashes establish ordering inside the versioned repository. They are not represented as an

externally registered pre-analysis plan. Frozen artifacts remain unchanged; second-round referee analyses are post-outcome exploratory.

The replication record contains the dated design protocol, definition of protected outcomes, information available at the freeze, hypothesis-to-test mapping, code, seeds, supports, and receipts. A second-round manifest connects every headline object to its machine-readable source:

TABLE 24. Headline-result lineage

Object	Repository result
Corrected baseline and SOC2 conditioning	<code>composition_influence/results/COMPOSITION_MODELS.csv</code>
Paired SOC2 movements	<code>composition_influence/results/COMPOSITION_PAIRED_DIFFERENCES.csv</code>
Influence and service exclusions	<code>composition_influence/results/JOINT_DELETION_AND_ROBUST_INFLUENCE.csv</code> ; <code>OCCUPATION_SERVICE_EXCLUSIONS.csv</code>
Canonical interval and architecture MDEs	<code>precision_rotation/results/CANONICAL_PRIMARY_INTERVAL.csv</code> ; <code>PAIRED_ARCHITECTURE_PRECISION.csv</code>
Quarterly, HAC, and pseudo-break results	<code>precision_rotation/results/QUARTERLY_ESTIMATES.csv</code> ; <code>ROTATION_TIME_HAC_SENSITIVITY.csv</code> ; <code>PSEUDO_BREAK_SUMMARY.json</code>
BCC grouping bridge	<code>bcc_bridge/results/BCC_GROUPING_ARCHITECTURE_RESULTS.csv</code> ; <code>BCC_GROUPING_PAIRED_DIFFERENCES.csv</code>
Bridge allocation and universe reconciliation	<code>bridge_uncertainty/results/MODEL_SENSITIVITIES.csv</code> ; <code>UNIVERSE_RECONCILIATION.csv</code>
Population-control audit	<code>population_controls/results/POPULATION_CONTROL_STATIC_SENSITIVITIES.csv</code>
Architecture spectrum and census	<code>architecture/results/ARCHITECTURE_EIGEN_SPECTRUM.csv</code> ; <code>framework_census/CANDIDATE_ARCHITECTURE_CENSUS.md</code>

Notes: All displayed paths are relative to `yax/revision/referee_round2_20260905/`; manifests in each directory record hashes. The mobility package is separately indexed in Appendix H.

Public-use CPS microdata are obtained from IPUMS and are not redistributed. Principal private-file hashes, not paths or credentials, appear in execution receipts. Scripts write aggregate results only. Every major workstream has a self-check or machine audit covering result identities, support counts, input/output hashes, and the absence of raw person identifiers.

The scientific article no longer uses “referee-requested,” “R1-style,” or “R2-style” to motivate an exercise. Revision chronology belongs here and in the response letter. The reason for each main-text analysis is stated in economic terms.

Author affiliation, email, acknowledgments, funding, conflicts, and final journal data-use language remain for author completion; none has been fabricated.